

Setting Up venv & Running the Streamlit App

Intel Image Classification — ML + ORB Project

1. What is a venv, and why use one?

A **virtual environment (venv)** is an isolated Python setup for this project only. It keeps this project's package versions (streamlit, opencv, scikit-learn, etc.) separate from other projects on your computer, avoiding version conflicts. It is never uploaded to GitHub — see the other guide's **.gitignore** section.

2. Create the Virtual Environment

Open a terminal inside your project folder (where app.py lives):

```
python -m venv venv
```

This creates a **venv/** folder containing an isolated Python interpreter and package storage for this project.

3. Activate the Virtual Environment

Windows (Command Prompt / PowerShell):

```
venv\Scripts\activate
```

macOS / Linux (bash/zsh):

```
source venv/bin/activate
```

You'll know it worked when your terminal prompt shows **(venv)** at the start of the line.

4. Install Requirements

With the venv active, install all dependencies in one step:

```
pip install -r requirements.txt
```

This reads requirements.txt and installs streamlit, opencv-python-headless, numpy, scikit-learn, joblib, and Pillow at the tested versions.

5. Run the App

Make sure **best_model.pkl** and **classes.json** are in the same folder as app.py, then run:

```
streamlit run app.py
```

Streamlit will print a local URL, typically **http://localhost:8501**, and should open automatically in your default browser. If it doesn't, copy the URL into your browser manually.

Using the app: click **Browse files**, choose a .jpg/.jpeg/.png image, and the predicted class + confidence will appear below it.

6. Stop / Close the App

1. Go back to the terminal window running Streamlit.
2. Press **Ctrl + C** (Windows/Linux) or **Cmd + C** (macOS) to stop the server.
3. Close the browser tab if it's still open.

7. Deactivate the Virtual Environment

When you're done working on the project, exit the venv with:

```
deactivate
```

This returns your terminal to the system-wide Python. Next time you work on the project, just re-activate the same venv (step 3) — no need to recreate it.

8. Quick Reference

Action	Command
Create venv	python -m venv venv
Activate (Windows)	venv\Scripts\activate
Activate (macOS/Linux)	source venv/bin/activate
Install packages	pip install -r requirements.txt
Run app	streamlit run app.py
Stop app	Ctrl + C in terminal
Deactivate venv	deactivate